

# **HOTEL BOOKING SYSTEM**

## **Java Mini Project Report**

**Name:** Mahek Yadav

**Roll no.:** 150096725142

**Team no.:** 7

**Cohort:** Sam Altman

## **1. ABSTRACT**

The Hotel Booking System is a Java Swing-based application developed to manage basic hotel room bookings. It allows users to book rooms, cancel bookings, check available rooms, and view all rooms with their booking status.

The project uses Java concepts such as classes, objects, arrays, LinkedList, HashMap, TreeMap, Swing GUI, event handling, and exception handling. It also calculates the total booking amount based on room price and number of days.

The main aim of this project is to provide a simple and user-friendly solution for basic hotel room management.

## **2. INTRODUCTION**

Managing hotel rooms manually can be time-consuming and may cause errors in booking and room availability.

The Hotel Booking System provides a simple desktop application for managing hotel rooms. Users can enter customer details, select a room, specify the number of days, and book or cancel a room.

The system uses Java Swing for the graphical interface and Java data structures for storing and managing information.

## **3. PROBLEM STATEMENT**

Hotels need to maintain information about rooms, customers, bookings, and room availability.

The proposed system solves this problem by providing features to:

- Book an available room
- Cancel a booking
- Check room availability
- Display all rooms

- Store customer and booking details
- Calculate the total booking amount

## **4. OBJECTIVES**

The main objectives of the project are:

- To develop a simple hotel booking application.
- To create a graphical interface using Java Swing.
- To manage rooms using arrays and collections.
- To implement room booking and cancellation.
- To search rooms using HashMap.
- To display rooms in sorted order using TreeMap.
- To handle invalid inputs using exception handling.

## **5. SCOPE OF THE PROJECT**

The project focuses on basic hotel room management.

The system allows users to:

- View available rooms
- Book rooms
- Cancel bookings
- View all rooms
- Enter customer details
- Calculate the total bill

The system does not include database storage, online booking, payment processing, or user authentication.

## **6. CONCEPTS USED**

### **6.1 Classes and Objects**

Classes are used to represent different entities of the system. The main classes are Room, Customer, Booking, and HotelBookingSystem.

### **6.2 Array**

An array is used to store the predefined hotel rooms.

### **6.3 LinkedList**

LinkedList is used to store booking records dynamically.

### **6.4 HashMap**

HashMap stores rooms using the room number as the key. It is used for quick room searching.

## **6.5 TreeMap**

TreeMap stores rooms in sorted order according to their room numbers.

## **6.6 Java Swing**

Java Swing is used to create the graphical user interface of the application.

# **7. SYSTEM REQUIREMENTS**

## **Hardware Requirements**

- Computer or Laptop
- Minimum 4 GB RAM

## **Software Requirements**

- Java JDK
- Java-supported IDE or VS Code
- Windows, macOS, or Linux

# **8. SYSTEM FUNCTIONALITIES**

## **Book Room**

The user enters customer details, room number, and number of days. The system checks the room availability and creates the booking.

## **Cancel Booking**

The user enters the room number. The system finds the booking and makes the room available again.

## **Available Rooms**

Displays all rooms that are currently available.

## **All Rooms**

Displays all rooms along with their type, price, and availability status.

## **Bill Calculation**

The total bill is calculated using:

Total Amount = Room Price × Number of Days

## **Input Validation**

The system checks for empty fields, invalid numbers, unavailable rooms, and invalid booking days.

# **9. SYSTEM DESIGN**

## **9.1 System Architecture**

User  
↓  
Java Swing GUI  
↓  
Hotel Booking System  
↓  
Room / Customer / Booking  
↓  
Java Data Structures

## **9.2 Class Structure**

Room

- Room Number
- Room Type
- Price
- Availability

Customer

- Name
- Phone Number

Booking

- Customer
- Room Number
- Number of Days
- Total Amount

HotelBookingSystem

- Room Array
- Booking LinkedList
- Room HashMap
- Sorted Room TreeMap

### 9.3 Data Flow

User enters details



System validates input



Room is searched



Booking or cancellation is performed



Room status is updated



Result is displayed

## 10. DATA STRUCTURE DESIGN

| Data Structure | Purpose                          |
|----------------|----------------------------------|
| Array          | Stores hotel rooms               |
| LinkedList     | Stores booking records           |
| HashMap        | Searches rooms using room number |
| TreeMap        | Displays rooms in sorted order   |

## 11. MODULE DESCRIPTION

### Room Management Module

Stores room number, room type, price, and availability status.

### Booking Module

Accepts customer details and creates a booking for an available room.

### Cancellation Module

Finds a booking using the room number and cancels it.

### Availability Module

Displays rooms that are currently available.

### Display Module

Displays all rooms along with their current status.

## 12. USER INTERFACE

The application contains the following components:

- Customer Name field
- Phone Number field
- Room Number field
- Number of Days field
- Book Room button
- Cancel Booking button
- Available Rooms button
- All Rooms button
- Output area

### Screenshots

Figure 1: Main Window

**Hotel Booking System**

**HOTEL BOOKING SYSTEM**  
Room Booking & Management

**Booking Details**

Customer Name:

Phone Number:

Room Number:

Number of Days:

**System Output**

Hotel Booking System

HOTEL BOOKING SYSTEM

Room Booking & Management

Booking Details

Customer Name:

Phone Number:

Room Number:

Number of Days:

Book Room

Cancel Booking

Available Rooms

All Rooms

System Output

===== BOOKING SUCCESSFUL =====

Customer : Mahek

Phone : 9382763524

Room : 301

Type : Deluxe

Days : 2

Price : ₹4000.0 / day

-----

TOTAL BILL: ₹8000.0

=====

Figure 3: Available Rooms

Hotel Booking System

# HOTEL BOOKING SYSTEM

Room Booking & Management

Booking Details

Customer Name:

Phone Number:

Room Number:

Number of Days:

Book Room

Cancel Booking

Available Rooms

All Rooms

System Output

===== AVAILABLE ROOMS =====

Room: 101 | Single | ₹1500.0/day

Room: 102 | Single | ₹1500.0/day

Room: 201 | Double | ₹2500.0/day

Room: 202 | Double | ₹2500.0/day

=====

Hotel Booking System

# HOTEL BOOKING SYSTEM

## Room Booking & Management

Booking Details

Customer Name:

Phone Number:

Room Number:

Number of Days:

Book Room

Cancel Booking

Available Rooms

All Rooms

System Output

===== ALL ROOMS =====

Room: 101 | Single | ₹1500.0 | Available

Room: 102 | Single | ₹1500.0 | Available

Room: 201 | Double | ₹2500.0 | Available

Room: 202 | Double | ₹2500.0 | Available

Room: 301 | Deluxe | ₹4000.0 | Booked

=====

Figure 5: Cancel Booking

The screenshot displays a Java Swing window titled "Hotel Booking System". The window has a title bar with three standard OS window control buttons (red, yellow, green). The main content area is titled "HOTEL BOOKING SYSTEM" in large, bold, black capital letters, with the subtitle "Room Booking & Management" centered below it. The interface is divided into several sections:

- Booking Details:** A section with four labels and corresponding text input fields: "Customer Name:", "Phone Number:", "Room Number:", and "Number of Days:". Each input field is a simple white rectangle with a thin grey border.
- Action Buttons:** Below the input fields, there are four buttons arranged in a 2x2 grid. The top row contains "Book Room" and "Cancel Booking". The bottom row contains "Available Rooms" and "All Rooms". All buttons are white with rounded corners and a thin grey border.
- System Output:** A section at the bottom of the window, titled "System Output", which contains a large text area. The text area displays the following output:

```
===== BOOKING CANCELLED =====  
Room      : 301  
Customer  : Mahek  
=====
```

## 13. IMPLEMENTATION

The project is implemented using Java and Java Swing.

The Room class stores room information, while the Customer class stores customer details. The Booking class stores booking information.

An array is used to store the predefined rooms. LinkedList stores booking records. HashMap is used to search rooms using room numbers, while TreeMap displays rooms in sorted order.

Swing components are used to create the GUI, and button actions are handled using event listeners.

## 14. EXCEPTION HANDLING AND VALIDATION

The system handles common input errors.

| Situation           | Message                                |
|---------------------|----------------------------------------|
| Empty fields        | Please fill all fields.                |
| Invalid numbers     | Room number and days must be numbers.  |
| Invalid days        | Number of days must be greater than 0. |
| Room not found      | Room not found.                        |
| Room already booked | Room is already booked.                |
| No booking found    | No booking found for this room.        |

## 15. RESULTS AND DISCUSSION

The Hotel Booking System successfully performs basic hotel room management.

The system can:

- Book rooms
- Cancel bookings
- Display available rooms
- Display all rooms
- Calculate booking bills
- Validate user input

HashMap allows rooms to be searched using their room numbers, while TreeMap keeps rooms sorted.

The Swing interface provides a simple way for users to interact with the system.

## 16. ADVANTAGES

- Simple and user-friendly interface
  - Easy room searching
  - Automatic bill calculation
  - Prevents booking of unavailable rooms
  - Uses multiple Java data structures
  - Provides input validation

## **17. LIMITATIONS**

- No database is used.
- Data is lost when the application closes.
- No login or authentication system.
- No online booking facility.
- Room details are predefined.

## **18. FUTURE ENHANCEMENTS**

The system can be further improved by adding:

- Database connectivity
- User login and authentication
- Online booking
- Payment integration
- Admin dashboard
- Booking history
- Report generation
- Web or mobile version

## **19. CONCLUSION**

The Hotel Booking System is a Java Swing application developed to manage basic hotel room operations. It provides features such as room booking, cancellation, availability checking, and bill calculation.

The project demonstrates important Java concepts including classes, objects, arrays, LinkedList, HashMap, TreeMap, Swing, event handling, and exception handling.

Overall, the project provides a simple solution for hotel room management and demonstrates how Java programming concepts can be used to develop a practical application.

## **20. REFERENCES**

1. Oracle Java Documentation
2. Java Swing Documentation
3. Java Collections Framework Documentation
4. Mini Project Documentation Guidelines provided by ITM Skills University